Question 1 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Three experiments are listed.

- I. Photoelectric effect
  II. Davisson-Germer experiment
  III. Young's double-slit experiment

Which of the experiments show the existence of the wave behavior of matter?

- A. II only
- B. II and III only
- C. I and III only
- D. I, II and III only

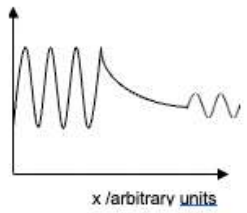
Question 2 Calculator Not Allowed

Markscheme

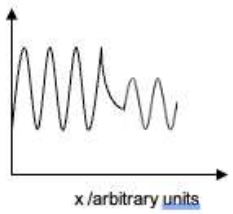
Difficulty: **Easy**

The graphs show the wavefunction of electrons that undergo tunneling through a barrier (reflected electrons are not considered). Which graph corresponds to the greatest width of barrier?

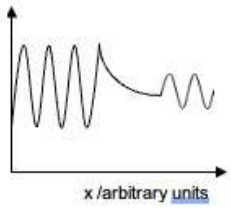
A.



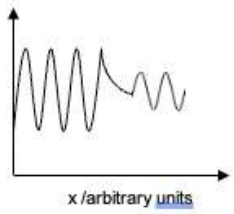
B.



C.



D.



Question 3 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following physical quantities has the same SI unit as the radioactive decay constant?

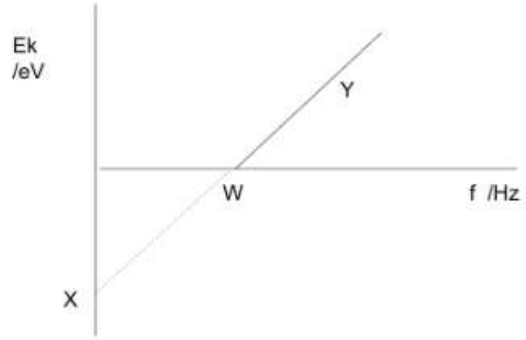
- A. Frequency
- B. Time period
- C. Wavelength
- D. Wave speed

Question 4 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The graph below shows the graph of the maximum kinetic energy (E_k)of the emitted electrons against incident frequency (f) plotted by a student investigating the work function for a particular metal.



Which of the following can be used by the student to get an approximate value for the work function of the metal in question?

- A. Intercept X
- B. Intercept W
- C. Gradient at Y
- D. $\frac{\text{Speed of Light}}{\text{Intercept W}}$

Question 5 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

In a vacuum chamber, electromagnetic radiation of frequencies $f_A < f_B$ just manage to produce a non-zero photocurrent from metals A and B respectively. When illuminated by an electromagnetic radiation of frequency $f > f_B$, the stopping voltage for the photocurrents from plates A and B are recorded to be V_A and V_B respectively.
Which of the following is true about V_A and V_B?

- A. $V_A < V_B$
- B. $V_A > V_B$
- C. $V_A = V_B$
- D. Information provided is insufficient to compare V_A and V_B

Question 6 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which of the following is not true about Bohr's model of the atom.

- A. It explains the stability of electrons within their energy levels.
- B. It correctly calculates the energies of electrons in all atoms.
- C. It explains the quantisation of the energy of hydrogen atoms.
- D. It assumes that the angular momentum of electrons is quantised.

Question 7 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Two lasers, A and B emit light of the same wavelength. If the intensity of A is greater than that of B, which of the following statements would be the correct interpretations of this situation using the wave and particle models of light.

	Wave model	Particle model
A.	The amplitude of LASER A is greater than that of LASER B.	LASER A emits more photons per second than LASER B.
B.	The amplitude of LASER B is greater than that of LASER A.	LASER B emits more photons per second than LASER A.
C.	The amplitude of LASER A is greater than that of LASER B.	LASER B emits more photons per second than LASER A.
D.	The amplitude of LASER B is greater than that of LASER A.	LASER A emits more photons per second than LASER B.

Question 8 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

A student investigates the scattering of alpha particles. Initially, she uses a gold foil (Z = 79) and next she repeats the experiment using an aluminum (Z = 13) foil of the same thickness. The scattered alpha particles are observed on a detector that fluoresces.
Which of the following will be true as she switches from the gold foil to the aluminum foil?

- I. The alpha particles deflect more from the gold nucleus than the aluminum.
  II. The alpha particles approach the gold nucleus closer than the aluminum nucleus.
  III. There is no change in the observed scattering pattern.

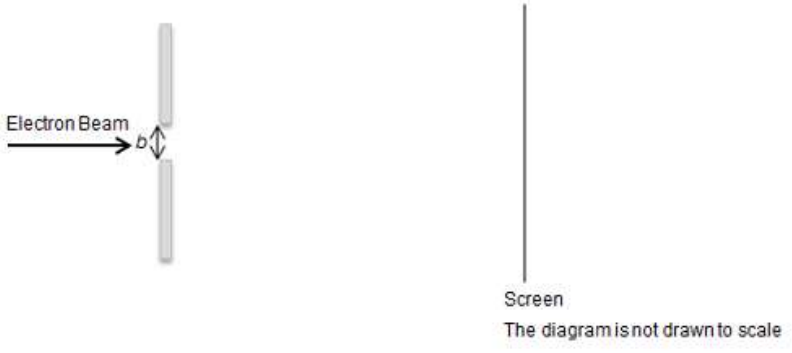
- A. I only
- B. II only
- C. III only
- D. II and III only

Question 9 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 6]
An electron beam is incident on a small circular hole of diameter b, and then observed on a screen as shown.



1. State what is meant by a particle's wave function. [1]
2. 1.

Describe the pattern that appeared on the screen [1]

2.
Explain the cause of the pattern described in b(i) above. [2]

3.
The hole diameter b is equal to $1.0 \times 10^{-10}\,m$.
Determine the uncertainty in the velocity of the electrons that is introduced when they pass through the slit. [2]

Question 10 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

Which value below gives the closest approximation for the diameter of a uranium-238 nucleus?

- A. $10^{-16}\,m$
- B. $10^{-14}\,m$
- C. $10^{-12}\,m$
- D. $10^{-10}\,m$

Question 11 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The nucleon numbers of elements X and Y are A and $8A$ respectively. Their radii are R_X and R_Y, and their densities are ρ_X and ρ_Y.

Which of the following shows the correct values of the ratios $\frac{R_Y}{R_X}$ and $\frac{\rho_Y}{\rho_X}$?

	$\frac{R_Y}{R_X}$	$\frac{\rho_Y}{\rho_X}$
A.	2	1
B.	8	2
C.	2	2
D.	8	1

Question 12 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

The activity of a 100% pure radioactive sample is $10^6\ Bq$. The activity of the sample after 20% of nuclei have decayed is:

- A. $0.2 \times 10^6\ Bq$
- B. $0.4 \times 10^6\ Bq$
- C. $0.6 \times 10^6\ Bq$
- D. $0.8 \times 10^6\ Bq$

Question 13 Calculator Not Allowed

Markscheme

Difficulty: **Easy**

During experiments observing beta decays, it was seen that the energy spectrum of the beta particles is continuous and momentum conservation was violated in beta decays. This provided evidence for the existence of:

- A. Nuclear energy levels
- B. Atomic energy levels

- C. Neutrinos
- D. Positrons

Question 14 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Light from a sodium lamp of wavelength is directed towards a metal surface and electrons until a stopping potential of V_s is applied. The wavelength of the incident light is halved. Which of the following corresponds to the new value of the required stopping potential?

- A. $\frac{V_s}{2}$
- B. $\frac{V_s}{2} - \frac{\phi}{e}$
- C. $2V_s$
- D. $2V_s + \frac{\phi}{e}$

Question 15 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

No electrons are emitted when metal plate X is irradiated by an electromagnetic radiation of wavelength λ, and intensity I. Which of the following could result in emission of photoelectrons?

- I. Decreasing the wavelength λ of incident light.
  II. Increasing the intensity of electromagnetic radiation at the same wavelength.
  III. Replacing metal X, with a metal of smaller threshold frequency.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

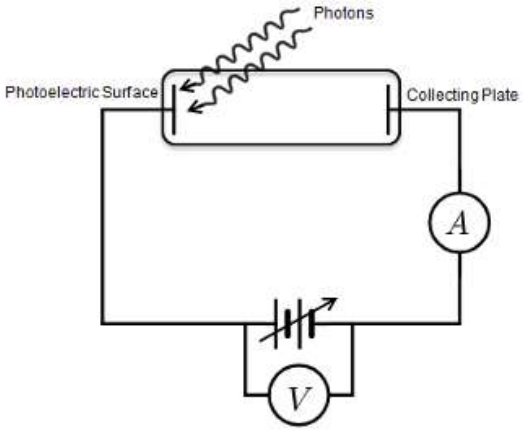
Question 16 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 12]

1. Light falls on a photoelectric surface with a threshold frequency of $5.2 \times 10^{14}\ Hz$.



[Source: Created with GeoGebra by Tom Walsh- https://www.geogebra.org/m/JbAKa82Y]

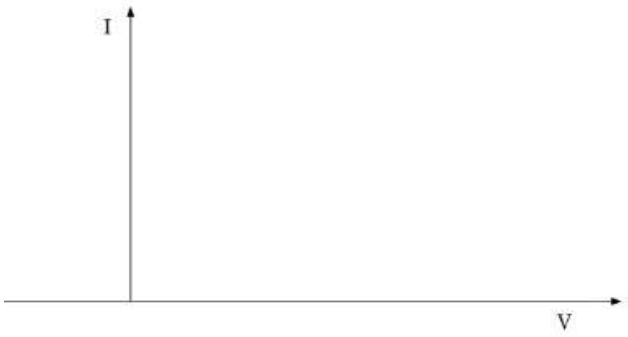
State what is meant by photoelectric emission. [1]

2.

1. Calculate the work function of the photoelectric surface in electronvolts. [2]
2. The ammeter gives a reading when a violet light is used, while it reads zero when a red light with the same intensity is used. Explain this observation. [3]
3. The frequency of violet photons is $7.5 \times 10^{14}\ Hz$. Calculate the maximum speed of the emitted electrons. [3]

3.

On the axes below, sketch the variation of the photoelectric current with the potential difference for an incident frequency above the threshold frequency. Also label the stopping voltage V_s. [3]

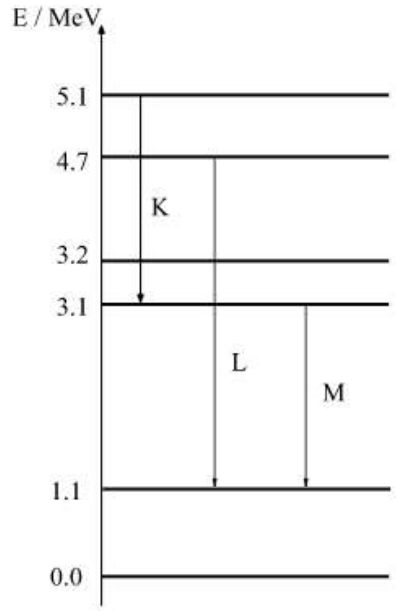


Question 17 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The nuclear energy levels of a nucleus are shown below. K, L, and M are transitions between energy levels.



Which statements are correct about photons emitted during transitions K, L, and M?

- I. The wavelength of the photon emitted by transition K is greater than the wavelength of the photon emitted by transition L.
  II. The frequency of the photon emitted by transition K is equal to the frequency of the photon emitted by transition M.
  III. The wavelength of the photon emitted by transition L is greater than the wavelength of the photon emitted by transition M.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II and III

Question 18 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Alpha particles of kinetic energy K are fired at nuclei of proton number Z.
The number of particles scattered at different angles are in agreement with the prediction of Rutherford's scattering model.
Which of the following changes to the values of K and Z will result in a significant deviation from Rutherford's model?

	Value of K	Value of Z
A.	Decrease	Increase
B.	Decrease	Decrease
C.	Increase	Increase
D.	Increase	Decrease

Question 19 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

A pure radioactive sample of mass m has a decay constant of λ. Its activity is given by A. Another pure sample of the same material has a mass $2m$. Consider the following statements about the sample of mass $2m$ after an interval of two half lives.

- I. The decay constant is $\frac{\lambda}{2}$.
  II. The activity is $\frac{A}{2}$.
  III. The mass of decayed nucleons in the sample is $\frac{m}{2}$.

Which of the statements is/are correct?

- A. I only
- B. II only
- C. III only
- D. II and III only

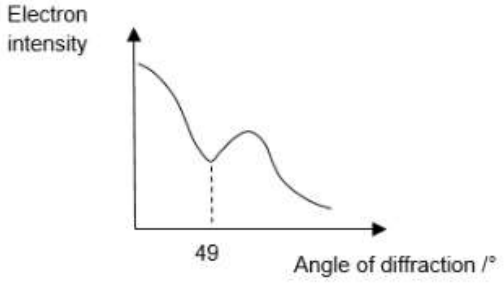
Question 20 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 10]

1. A beam of electrons is incident on an iron foil ($Z = 56$) and electrons are diffracted. In the experiment, electrons have initial kinetic energy of $42\,GeV$. The first diffraction minimum is observed at an angle of 49° from the average position of most of the electrons detected.



   1. Calculate the density of the iron nucleus. [3]
   2. Using the diffraction information given, estimate the radius r of the iron nucleus . [4]
2. In another experiment, alpha particles are emitted towards iron foil with an initial kinetic energy of $125\,keV$. Estimate the distance of the closest approach to the iron nucleus of the alpha particles. [3]

Question 21 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 12]

1. Define decay constant. [1]
2. Mushrooms contain a radioactive element called caesium-137, $^{137}_{55}\mathrm{Cs}$. The element has a half-life of $30\,yr$ and a sample of living mushrooms has a constant decay rate of $2000\,Bq$ for every kg of mass. A sample of $50\,g$ is extracted from a dried mushroom and the activity is measured as $0.80\,Bq$.
   1.

Caesium-137 decays to barium-137, $^{137}_{56}\text{Ba}$, by beta emission. Complete the nuclear reaction equation of this decay, showing both particles emitted in the decay.

$^{137}_{55}\text{Cs} \rightarrow$ _____ + $^{137}_{56}\text{Ba}$ + _____ [2]

2.

Show that the decay constant of caesium-137 is about $0.023\,yr^{-1}$ [2]

3.

Given that the amount of caesium in the mushroom stops being replenished once the mushroom is no longer living, determine the age of the dried mushroom. [3]

3.

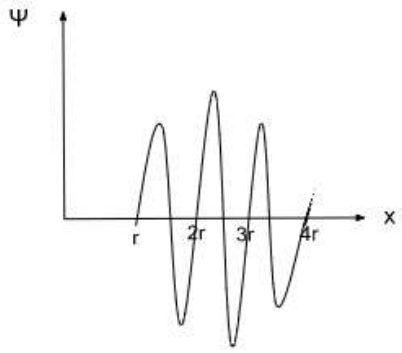
Some other radioactive substances have half-lives of thousands of years. Outline the method used for determining the half-life of these kinds of substances. [4]

Question 22 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The variation of the wave function Ψ of an electron with its distance to nucleus x is shown in the graph below.



Which one of the following gives the uncertainty in the momentum of this electron?

- A. $\frac{h}{4\pi r}$
- B. $\frac{h}{8\pi r}$
- C. $\frac{h}{12\pi r}$
- D. $\frac{h}{16\pi r}$

Question 23 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

The photocurrents produced by illuminating a metal plate with yellow light beams of intensities I_1 and I_2 are i_1 and i_2 respectively. The corresponding stopping voltages are V_1 and V_2. Given that $I_1 < I_2$, which of the following statements is true about the photocurrents and the stopping voltages?

	Photocurrents	**Stopping voltages**
A.	$i_1 < i_2$	$V_1 = V_2$
B.	$i_1 < i_2$	$V_1 < V_2$
C.	$i_1 > i_2$	$V_1 = V_2$
D.	$i_1 = i_2$	$V_1 < V_2$

Question 24 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 9]

1. De Broglie related the wavelength of a particle to its momentum. State what is meant by matter waves. [1]
2.
    1. Show that the wavelength λ of an electron of mass m and kinetic energy E can be found from [2]
        - 2. $\lambda = \frac{h}{\sqrt{2mE}}$
    1. Assuming that the magnitude of the energy of a given energy level of an atom is equal to the kinetic energy of the electron, show that the wavelength of the electron in the second energy level in a hydrogen atom is approximately $6.7 \times 10^{-10}\, m$. [3]
    2. Use the wavelength from b(ii) to estimate the orbital radius of this electron. [3]

Question 25 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum marks: 9]

1. A thorium-234 ($\mathrm{Th\text{-}234}$) nucleus undergoes beta decay and transforms into a protactinium-234 nucleus ($\mathrm{Pa\text{-}234}$). The binding energies per nucleon for Thorium-234 and protactinium-234 are $7.84\ \ MeV$ and $7.92\ \ MeV$ respectively. An electron has a rest mass energy of $0.020\ MeV$.
The reaction can be represented as follows:
$^{234}_{90}\mathrm{Th} \rightarrow\ ^{234}_{91}\mathrm{Pa} + 10e^- + \overline{\nu_e}$
    1. Explain what is meant by binding energy per nucleon. [2]
    2. Calculate, in joules, the total energy released in the decay. [3]
    3. State the name of the exchange particle in the reaction. [1]
2. A cubic container is filled with pure $\mathrm{thorium\text{-}234}$. The container does not contain any other elements. After 62 days, the ratio of the number of nuclei of decay product to $\mathrm{thorium\text{-}234}$ nuclei is 5.
Calculate, in days, the half-life of the $\mathrm{thorium\text{-}234}$ nuclei. [3]

Question 26 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Which of the following is correct for an electron neutrino?

- A. An electron neutrino has a continuous distribution of energies when produced in radioactive decay.
- B. The charge of an electron neutrino is $1.6 \times 10^{-19}\, C$.
- C. The speed of an electron neutrino is $3.0 \times 10^{8}\, m\, s^{-1}$.
- D. Electron neutrinos are produced during alpha decays.

Question 27 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

An electron with a kinetic energy E_k annihilates with a positron that moves in the opposite direction with the same kinetic energy. In this process, two photons are produced with a wavelength of λ and momentum of p. Which of the following is correct for produced photons when the kinetic energy of colliding particles is greater than E_k?

	Wavelength	Momentum
A.	Greater than λ	Greater than p
B.	Greater than λ	Less than p
C.	Less than λ	Greater than p
D.	Less than λ	Less than p

Question 28 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

Red and blue light beams of equal intensities illuminate a photosensitive plate and produce photocurrents I_R and I_B respectively. The respective stopping voltages against the two photocurrents are V_R and V_B.
Consider the following statements:

- I. $I_R > I_B$
  II. $V_R < V_B$
  III. Photoelectrons with higher kinetic energy result in a higher current.

Which statements are correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

Question 29 Calculator Not Allowed

Markscheme

Difficulty: **Medium**

In a laboratory experiment, n moles of photons of wavelength λ are normally incident every second per square meter on a solar sail. Which of the following expressions correctly states the photo-pressure on the solar sail, given that the photons are reflected elastically.

- A. $\frac{h}{\lambda}$
- B. $\frac{2h}{\lambda}$
- C. $(n)(N_A)\frac{h}{\lambda}$
- D. $(n)(N_A)\frac{2h}{\lambda}$

Question 30 Calculator Not Allowed

Markscheme

Difficulty: **Hard**

The uncertainty in the position of a proton is $10^{-10}\,m$, and its speed is $7 \times 10^{5}\,m\,s^{-1}$. What is the fractional uncertainty in its speed?

- A. 5×10^{-4}
- B. 5×10^{2}
- C. 5×10^{-8}
- D. 2×10^{-4}

Question 31 Calculator Not Allowed

Markscheme

Difficulty: **Hard**

Two photosensitive plates A and B with threshold frequencies f_A and f_B are illuminated in a vacuum with an electromagnetic radiation of wavelength λ such that photoelectrons are emitted from both plates. The corresponding stopping voltages are recorded to be V_A and V_B respectively.

The ratio $\dfrac{V_B - V_A}{f_A - f_B}$ is an estimate of:

- A. hc
- B. $\dfrac{hc}{\lambda}$
- C. h
- D. $\dfrac{h}{e}$

Question 32 Calculator Not Allowed

Markscheme

Difficulty: **Hard**

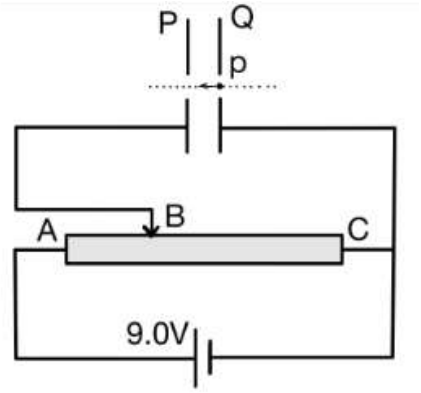
An electron is incident normally on a narrow rectangular opening of length l and width w. After passing through the slit, the electron's momentum has a net component p in the plane of the rectangular slit. The minimum uncertainty in p is:

- A. proportional to $\dfrac{1}{l} + \dfrac{1}{w}$
- B. proportional to $\dfrac{1}{\sqrt{lw}}$
- C. proportional to $\sqrt{\dfrac{1}{l^2} + \dfrac{1}{w^2}}$
- D. proportional to $\sqrt{\dfrac{1}{l^2 + w^2}}$

Question 33 Calculator Not Allowed

Markscheme

Difficulty: **Hard**



In the schematic diagram above, P and Q are a pair of charged plates connected to a potential divider across a $9.0\ V$ source. The sliding contact is placed such that $BC = 2AB$.

When protons, p of kinetic energy $4.5\ eV$, are shot from the center of the hole in plate Q, some are detected on the left of plate P due to quantum effects. Which of the following changes would increase the probability of the particle appearing on the left side of plate P?

- I. Replacing the proton with a positron of the same energy.
  II. Bringing plate P closer to Q.
  III. Sliding the pointer of the potential divider from B towards C.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III